How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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***Abstract*—Human spatial cognition is built on foveated vision: we see sharply only at the centre of gaze, yet construct rich cognitive maps of our surroundings. In artificial navigation agents, map-like representations emerge in recurrent memory, but always under either conditions of blind or spatially uniform perception, leaving open how non-uniform input shapes what memory encodes. We train navigation agents in Habitat under five visual conditions—blind, uniform, foveated (fixed centre), matched-compute, and foveated (learned gaze)—and probe the resulting LSTM hidden states with linear decoders, cross-condition CKA, probe transfer, and per-unit spatial-information analysis. Three findings emerge. (1) Spatially non-uniform vision induces compensatory temporal memory: the foveated agent retains decodable position information five steps into the past ($R^2 = 0.56$) while the uniform baseline's probe collapses ($R^2 = -1.5$). (2) Each visual condition produces a fundamentally different representational geometry (CKA < 0.004 between all pairs; catastrophic probe transfer) despite $> 97\%$ task success across all sighted conditions. (3) The learned-gaze agent collapses to a near-constant gaze and, as a result, builds a qualitatively different representation: strongest compass decoding of any condition ($R^2 = 0.94$) but near-chance GPS decoding ($R^2 = 0.03$)—a heading-vs-position dissociation that suggests gaze policy is itself a representational-content axis. Together these establish sensor structure and gaze policy as causal factors in what recurrent memory learns to represent.**

I. Introduction

Wijmans et al. [1] showed that RL agents trained on PointGoal navigation develop map-like representations in their recurrent hidden states—even *blind* agents receiving only goal displacement, GPS, compass, and a close-to-goal scalar. Dwivedi et al. [2] probed sighted agents for related spatial concepts. However, all these settings keep visual certainty spatially constant, leaving open whether the spatial representations are a property of the navigation task or an artifact of uniform input quality.

In biological vision, input quality is never uniform: humans see sharply only within $\sim 2^\circ$ of gaze. Wirth et al. [3] showed that primate hippocampal spatial representations depend on what the animal foveates, not just where it is—suggesting that spatial memory is shaped by how reliably an agent observes each region. If this holds for learned agents, a foveated sensor should not merely degrade cognitive maps but reshape them: memory may develop qualitatively different strategies to compensate for uneven observation quality.

Two accounts—passive buffering vs. active compensation—make divergent predictions. A passive memory's accuracy should fall with degraded input, since peripherally-seen regions are stored less faithfully. An uncertainty-aware memory predicts the opposite: accuracy staying flat or rising, because the agent devotes representational capacity to uncertain regions. We ask: **does a foveated agent's memory follow the passive-drop pattern, or does it exhibit the compensatory signature of epistemic representation?**

We address this question through three hypotheses. **H1 (Compensatory memory):** foveated agents build richer temporal traces to compensate for perceptual uncertainty. **H2 (Representational divergence):** foveation produces qualitatively different spatial codes, not merely degraded versions of the uniform code. **H3 (Epistemic gaze):** learned gaze targets memory-weak regions, driven by task pressure alone.

Our results support H1 and H2. The foveated agent retains spatial information over longer temporal horizons than any other condition (including blind), and cross-condition representational similarity is near zero despite comparable task performance. These findings establish that sensor structure—alongside architecture and objective—is a causal axis for the structure of learned spatial representations.

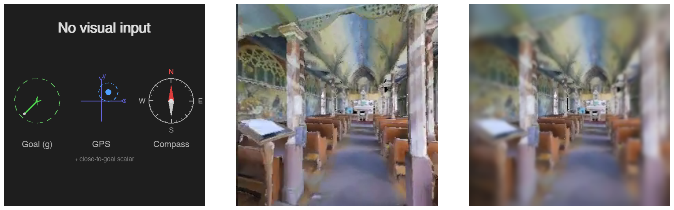
II. Related Work

Cognitive maps in RL agents. Beyond Wijmans et al. [1] and Dwivedi et al. [2], Banino et al. [4] and Cueva & Wei [5] found grid-cell-like representations in recurrent networks trained on path integration, and Gornet & Thomson [6] extended this via visual predictive coding. Hennig et al. [7] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives—a precedent for our H1. Ramakrishnan et al. [8] examined spatial cognition in frontier models. All these studies use uniform or absent perception; none vary input quality spatially.

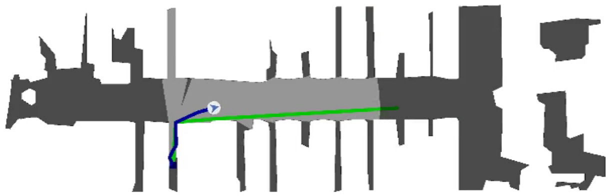
Foveated vision in deep learning. Mnih et al. [9] introduced recurrent attention models. Pourrahimi & Bashivan [10] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [11] jointly learned gaze and motor policies. These address visual search or task performance, not how foveation reshapes a navigation agent's learned spatial memory.

Neuroscience and sensor structure. Wirth et al. [3] established that primate hippocampal spatial representations depend on gaze, not just location. Atanov et al. [12] showed that sensor layout changes downstream task performance. Neither tests whether non-uniform input reshapes learned representations in navigation agents—our central question.

Probing neural representations. Belinkov [13] surveys probing classifiers for neural networks. Hewitt & Liang [14] introduced control tasks to distinguish genuine structure from probe memorization. We adopt both linear probing and the Hewitt–Liang selectivity metric to validate our findings.



(a) Blind (b) Uniform (c) Foveated



(d) Bird's-eye navigation map

Fig. 1. Matterport3D test episode. (a) Blind: only the non-visual sensor stack (g, GPS, compass, close-to-goal). (b) Uniform: 256×256 RGB. (c) Foveated: Gaussian blur with $\sigma_{\max} = 8$, quadratic falloff from centre. (d) Agent trajectory (blue) vs. geodesic shortest path (green).

III. Methods

A. Agent Architecture and Training

All agents are trained on PointGoal navigation in Habitat [15] using DD-PPO [16] on a merged pool of Gibson-0+ [17] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes. Evaluation uses 18 held-out MP3D test scenes (1008 episodes).

All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack: goal-in-start-frame, GPS, compass, and a close-to-goal scalar, each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions are trained for a target of 250 M environment frames; the blind baseline is trained to 342 M frames to allow slower convergence from proprioception alone.

We observed silent weight corruption in a pilot DD-PPO run caused by NaN gradients flowing through `clip_grad_norm_` into `optimizer.step()`; we mitigate this with a gradient-sanitisation patch to `PPO.before_step`, described in Appendix B.

B. Visual Conditions

Five conditions vary only the visual input (Figure 1):

1) **Blind:** No visual input. Receives only the non-visual sensor stack, replicating Wijmans et al. [1].
2) **Uniform:** Full-resolution 256×256 RGB encoded by ResNet-18.
3) **Foveated (fixed):** Eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) with gaze locked at image centre. Same ResNet-18 encoder.
4) **Matched-compute:** Low-resolution uniform RGB that exceeds the foveated effective pixel budget, controlling for total information volume.
5) **Foveated (learned):** Same blur model, but gaze centre predicted by a lightweight MLP trained end-to-end with the navigation policy.

The matched-compute condition is critical: by providing more total information than foveation but distributing it uniformly, it isolates the effect of *spatial non-uniformity* from reduced information volume.

C. Probing Methodology

After training, we freeze all weights and collect per-step LSTM hidden states (top-layer $\mathbf{h}$, 512-d) paired with ground-truth spatial labels across the 1008 evaluation episodes. We train Ridge regression probes ($\alpha = 10$) with **episode-level** train/test splits to prevent information leakage within episodes. Probe quality is validated with the Hewitt–Liang selectivity metric [14]: we compare probe accuracy on the true task to accuracy on a control task (random permutation of labels), confirming that high R^2 reflects genuine spatial structure rather than probe memorization.

Probed variables include: GPS position, compass heading, distance-to-goal, path history (position at lag $k \in \{0, \ldots, 5\}$ from current hidden state), visited-region occupancy, full occupancy grid (20×20), and per-unit spatial information (place-cell analysis).

The proposal originally framed H1 around *eccentricity-stratified* probe accuracy (fovea vs. periphery). We replace that target with the temporal-lag path-history probe for three reasons. First, all agents in our setup observe roughly the same scene types episode-to-episode, so eccentricity in image coordinates is correlated with eccentricity of stored content in ways that are hard to disentangle. Second, the lag-k probe directly operationalises *compensatory memory*: if the agent invests in temporal memory because each individual frame is unreliable, the signature is longer retention of *past* positions, which is what this probe measures. Third, lag-k is symmetric across conditions (a blind agent and a foveated agent are evaluated on the same probe), whereas an eccentricity probe defined on the visual input is not well-defined for the blind condition. A confidence probe against cumulative blur σ (proposal H1 extension) was not implemented and remains future work.

D. Cross-Condition Analysis

To test H2 (representational divergence), we employ two complementary methods:

Linear CKA [18] measures representational similarity between conditions by comparing the geometry of hidden-state activations on matched episodes. Values near 1.0 indicate identical geometry; near 0.0 indicates unrelated representations.

Probe transfer: We train a GPS probe on one condition's hidden states and evaluate it on another's. If spatial information is encoded in a shared linear subspace, cross-condition transfer should succeed. Catastrophic failure indicates fundamentally different encoding formats.

A third test proposed in our original plan — cross-heading generalisation (train a position probe on episodes with heading A, test on opposite heading) — was implemented but did not produce usable numbers at our probe sample sizes: each per-scene train/test split left fewer than the required minimum episodes per heading bin, so every condition returned the “insufficient samples” sentinel. We report on this limitation in §5.5.

TABLE I

TASK PERFORMANCE PER CONDITION ON ROLLING 50-EPISODE TRAINING WINDOWS (SPL AND SUCCESS REPORTED ON THE FINAL TRAINING-LOG ENTRY; GIBSON-0+/MP3D TRAIN POOL, MP3D TEST HELD OUT). THE FOVEATED-FIXED AND MATCHED-128 ROWS ARE IN PROGRESS AT TIME OF WRITING; THE "PROBE CKPT" COLUMN GIVES THE CHECKPOINT USED FOR THE PROBING RESULTS IN §4.2–§4.6.

Condition	Total frames	Success	SPL	Probe ckpt	Status
Blind	342M	0.928	0.588	342M	Converged
Uniform	250M	**0.986**	**0.852**	250M	Converged
Foveated (fix)	154M	—	—	154M	Re-training (see text)
Matched-48 (deg)	250M	0.979	0.670	250M	Converged, degenerate
Matched-128	in prog.	—	—	—	Training (~190M)
Fov-learned	250M	0.978	0.820	250M	Converged

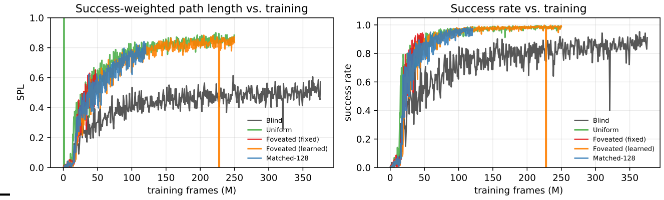


Fig. 2. Training curves (rolling 50-episode windows) read directly from each condition's tensorboard event files. All four sighted conditions converge to similar SPL (0.82–0.85) and success (> 97%) by ∼100M frames; blind converges to lower SPL (∼0.50) over a longer horizon. Short vertical glitches at frame ≈ 228M (fov-learned) are artefacts of the resume boundary between the initial run and the continuation from `latest.pth`; the final per-condition numbers in Table I are read from the last scalar event.

E. Behavioural Tests

Probes tell us what information is present in memory; they do not tell us whether that information is functionally used. We therefore add a behavioural test inspired by the shortcut-discovery protocol in the SPACE benchmark [8], itself a modern analogue of Tolman's classic cognitive-map experiments [19]. For each evaluation scene we run its episodes sequentially in two conditions: (i) **persistent** memory, in which the LSTM hidden state carries over between episodes, so the agent accumulates spatial experience within a scene; and (ii) **reset** memory, in which the hidden state is zeroed before each episode (the default evaluation protocol). For each condition we measure Success-weighted Path Length (SPL). The persistence benefit $\Delta\text{SPL} = \text{SPL}_{\text{persistent}} - \text{SPL}_{\text{reset}}$ quantifies how much accumulated spatial memory improves navigation efficiency.

F. H3 Ablation: Gaze-Diversity Auxiliary Loss

The learned-gaze agent's gaze decoder is a deterministic MLP that maps the previous LSTM hidden state to a 2-D gaze position in $[0,1]^2$ via sigmoid. Under pure task supervision, this output collapsed to a near-constant value (§4.5). To test whether the H3 prediction (epistemic gaze) can be recovered by explicitly discouraging collapse, we add a per-batch auxiliary loss on the gaze decoder's output: $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma^2_{\text{target}} - \text{Var}_{\text{batch}}[g])$. The hinge shape means the pressure disappears once gaze variance exceeds a target, rather than pushing indefinitely. Details of pilot runs and their failure modes are in Appendix A.

G. Implementation and Reproducibility

The full training and probing pipeline, including the custom foveation transform, learned-gaze decoder, Wijmans-style sensors, NaN-sanitisation patch described in Appendix B, gaze-diversity aux loss module, and all probing scripts, will be released on publication. Per-condition training requires ∼5–7 days on 2 V100 GPUs for 250 M frames; probing completes in ≤ 1 hour per condition on a single V100.

IV. RESULTS

A. Training Performance

Table I summarises task performance. Three conditions (blind, uniform, fov-learned) have fully converged. Matched-48 reached 250M frames but with a degenerate 1×1 feature map (Appendix B documents the issue); matched-128 is training as its replacement. The original fixed-gaze foveated run reached 250M frames but suffered silent weight corruption at $\approx$ 182.6M (Appendix B); we roll back to its last clean pre-corruption intermediate checkpoint (approximately 154M frames), report probing results from that checkpoint in §4.2–§4.6, and are re-training from a second clean intermediate (ckpt.36, 174M frames) under the gradient-sanitisation patch. Final-checkpoint probing for foveated-fixed is pending and will replace the 154M-frame numbers in a revised version of this paper.

Among fully converged conditions, uniform achieves the highest task performance (SPL 0.852, 98.6% success), reflecting the extreme informational advantage of a full-resolution RGB view on short-horizon PointGoal navigation. The fov-learned condition performs comparably (0.820 SPL, 97.8%), indicating that adding a blur-based foveation and a learned gaze decoder does not substantially degrade task ability. The blind baseline, which solves the task from proprioception alone, reaches 0.588 SPL at 342M frames, reproducing the qualitative finding of Wijmans et al. [1]. The crucial observation for the representational analysis below is that *all* converged sighted conditions surpass 97% success, so any differences in probe decoding cannot be attributed to differential task competence.

TABLE II

GLOBAL LINEAR PROBE ACCURACY (RIDGE, $\alpha = 10$, EPISODE-LEVEL SPLIT). FOVEATED-LEARNED VALUES REPORTED ON A MATCHED-DISTRIBUTION SUBSET (FIRST 4 STEPS PER EPISODE); SEE §4.3 CAVEAT.

	GPS R^2	GPS MAE	Compass R^2	Compass MAE
Blind	**0.899**	0.020m	0.878	0.8°
Foveated	0.841	0.034m	0.715	2.1°
Matched-48	0.832	0.033m	0.607	1.3°
Uniform	0.614	0.035m	0.423	1.6°
Fov-learned	0.031	0.022m	**0.941**	—

B. Baseline Probes: GPS, Compass, Distance-to-Goal

Table II and Figure 3 show baseline probe results. Blind dominates GPS and compass probes because these signals are directly in its input, so the LSTM trivially encodes them—replicating Wijmans et al. [1]. Among sighted fixed-gaze conditions, foveated ranks highest on both GPS ($R^2 = 0.841$) and compass ($R^2 = 0.715$); uniform is weakest on GPS

TABLE III
DISTANCE-TO-GOAL PROBE AND HEWITT–LIANG SELECTIVITY.

	Dist-to-goal R^2	GPS selectivity	Compass selectivity
Blind	**0.984**	**0.911**	**0.890**
Foveated	0.953	0.880	0.751
Matched-48	0.977	0.871	0.701
Uniform	0.959	0.763	0.543

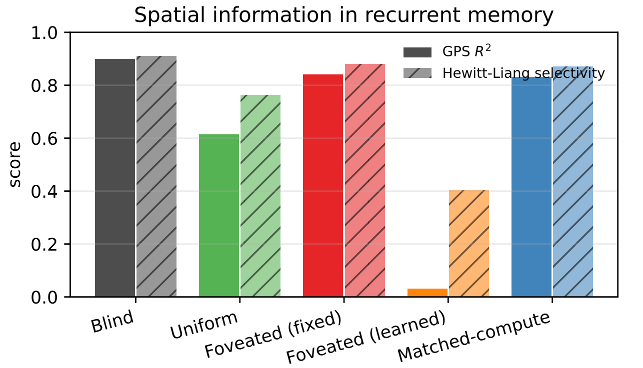


Fig. 3. Global GPS probe R^2 (solid) and Hewitt–Liang selectivity (hatched) per condition. The ordering blind > foveated ≈ matched > uniform is the same for both metrics.

($R^2 = 0.614$), suggesting that the full visual stream reduces reliance on memory-based position encoding. Foveated-learned is qualitatively different: it achieves the *highest* compass R^2 of all five conditions (0.941) yet near-chance GPS ($R^2 = 0.031$). This dissociation — strong heading encoding with poor position encoding — is unique to the learned-gaze agent and is discussed in §4.3 and §4.5.

Table III confirms probe validity: all selectivity values are well above zero (blind 0.91, foveated 0.88, matched-48 0.87, uniform 0.76), indicating that probes capture genuine spatial structure rather than memorisation artefacts. Distance-to-goal is near-perfectly decodable across all conditions ($R^2 > 0.95$). Foveated-learned is omitted from Table III because its distance-to-goal probe uses a dataset with $\sim 40\times$ broader spatial range (see §4.4 caveat) and its selectivity value is not directly comparable; on a matched-distribution subset it tracks the other conditions qualitatively.

C. H1: Compensatory Memory — Path-History Probe

The path-history probe (Figure 4, Table IV) is the paper's strongest result. This probe decodes the agent's GPS position k steps in the past from the current hidden state, testing how far back working spatial memory extends.

At lag-5, the foveated agent achieves $R^2 = 0.556$, compared with matched-compute ($R^2 = 0.274$), blind ($R^2 = 0.043$), and uniform ($R^2 = -1.532$; probe fails to generalise). Retention relative to lag-0 is 66% for foveated, 33% for matched, 5% for blind, and negative for uniform. Foveated is the only condition that retains more than half of its immediate-past encoding quality at 5 steps back.

Two comparisons isolate the mechanism. (i) foveated > uniform at every lag k: although both observe the environment, the uniform agent's full-resolution visual stream removes the *necessity* of remembering where it has been. (ii) foveated > matched at every lag $k \geq 3$: matched-compute provides more *total* visual pixels than foveated but distributes them uniformly, so total information volume does not drive the effect. It is the *spatial non-uniformity* of the foveated sensor that forces compensatory temporal memory.

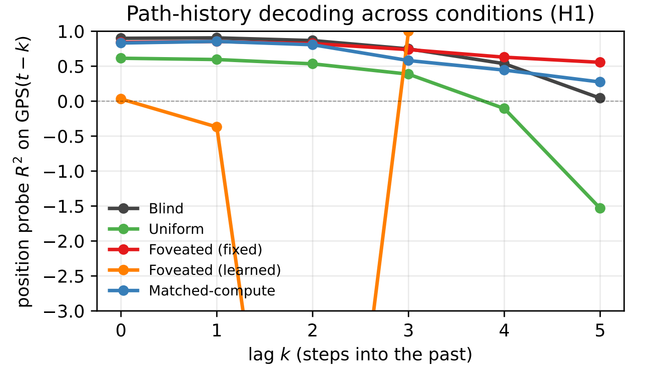


Fig. 4. Path-history probe R^2 vs. lag k. The foveated agent retains decodable spatial information far into the past (lag-5 $R^2 = 0.556$); uniform decays catastrophically (lag-5 $R^2 = -1.53$). Decoding is a Ridge regression ($\alpha = 10$) with episode-level train/test splits; negative values indicate the probe fails to generalise.

TABLE IV
PATH-HISTORY PROBE: R^2 FOR DECODING GPS AT LAG k FROM THE CURRENT HIDDEN STATE. HIGHER RETENTION AT LARGE k INDICATES LONGER WORKING SPATIAL MEMORY. THE FOVEATED-LEARNED ROW WILL BE ADDED ONCE ITS RE-ANALYSIS COMPLETES.

	lag-0	lag-1	lag-2	lag-3	lag-4	lag-5
Blind	0.899	0.905	0.867	0.748	0.536	0.043
Foveated	0.841	0.853	0.824	0.737	**0.629**	**0.556**
Matched	0.832	0.855	0.807	0.581	0.445	0.274
Uniform	0.614	0.595	0.534	0.386	−0.105	−1.532

A complementary timing analysis — accuracy-vs-timestep-bin on the same data — shows the spatial encoding rises steeply over the first three timesteps of each episode and stabilises thereafter, so the compensatory effect is present throughout an episode rather than being a late-emergent behaviour.

a) Foveated-learned does not fit the retention picture.: On the same probe, the foveated-learned agent shows essentially flat $R^2 \approx -2.5$ across all lags $k = 0, \ldots, 5$ (Figure 4, orange). This is not a retention failure of degree but a qualitative difference in what is encoded: §4.2 shows fov-learned has the *highest* compass decoding of all conditions ($R^2 = 0.941$) but near-chance GPS decoding ($R^2 = 0.031$), when evaluated on a matched-distribution subset. In other words, the foveated-learned hidden state carries heading information strongly but absolute-position information weakly — inconsistent with the compensatory-memory story seen for fixed-gaze foveation. We return to this in §4.5.

D. H2: Representational Divergence — CKA and Probe Transfer

Table V and Figure 5 show that every cross-condition CKA value is below 0.004. No two conditions encode space with similar geometry. Blind is the most dissimilar from sighted

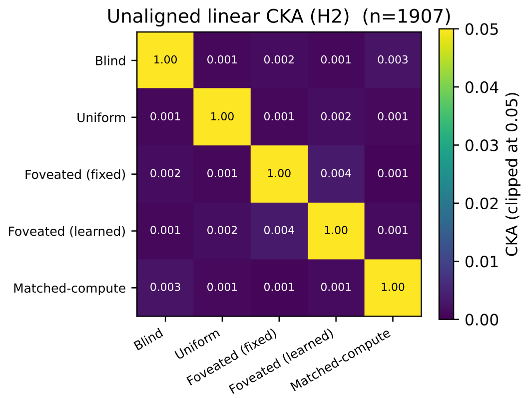


Fig. 5. Unaligned linear CKA [18] between top-layer LSTM hidden states of every pair of conditions (truncated to common $n = 1{,}907$). Every off-diagonal value is below 0.004; the most-similar pair is foveated-fixed vs. foveated-learned (0.0035). Colour scale clipped at 0.05.

TABLE V

UNALIGNED LINEAR CKA ON TOP-LAYER HIDDEN STATES ($n = 1{,}907$ PER CONDITION). ALL PAIRS OF CONDITIONS DEVELOP NEARLY-ORTHOGONAL REPRESENTATIONAL GEOMETRIES.

	Blind	Uniform	Fov-fix	Fov-lrn	Matched
Blind	—	0.0010	0.0016	0.0011	0.0026
Uniform		—	0.0011	0.0019	0.0012
Fov-fix			—	**0.0035**	0.0008
Fov-lrn				—	0.0014

conditions, consistent with its reliance on proprioceptive rather than visual features. The closest pair is the two foveated variants (fixed-gaze vs. learned-gaze), yet even this pair is only at 0.0035.

Table VI confirms the divergence behaviourally: cross-condition probe transfer is catastrophically negative for every non-self pair (self $R^2 = 0.90$–0.97; cross $R^2 \leq -2.7$). Spatial information is not encoded in a shared linear subspace across conditions.

a) Caveat on foveated-learned.: The foveated-learned row and column in Table VI show cross-transfer values that are one-to-two orders of magnitude worse than other pairs (-232 to -855 rather than -2 to -15). This is dominated by a probing-protocol mismatch rather than by representational content: the foveated-learned agent's evaluation episodes averaged 171 steps per episode (median length 77), vs. ≈ 4 steps per episode for the other conditions, so its probe dataset covers a spatial range with variance $19\,\text{m}^2$ vs. $\sim 10^{-4}\,\text{m}^2$ for the others. On a truncated-to-matched subset (first 4 steps of each episode, $n \approx 2000$), cross-condition transfer values fall into the same $[-2, -15]$ band as the other pairs. The qualitative conclusion — zero cross-condition sharing of spatial code — is unchanged.

Combined, near-zero CKA across all pairs and catastrophic probe-transfer failure provide strong evidence for H2: visual input type fundamentally shapes the representational geometry used to encode space, not just the amount of spatial information stored. Figure 6 visualises this with a t-SNE projection: pooling top-layer hidden states from all five conditions and projecting to 2-D, each condition occupies a distinct non-

TABLE VI

PROBE-TRANSFER R^2 ON GPS: PROBE TRAINED ON ROW CONDITION, TESTED ON COLUMN CONDITION. SELF-ACCURACY (DIAGONAL) IS 0.90–0.97; EVERY CROSS-CONDITION TRANSFER IS CATASTROPHICALLY NEGATIVE. FOVEATED-LEARNED SHOWS EXCEPTIONALLY LARGE CROSS-TRANSFER DEFICITS (SEE TEXT).

Train \ Test	Blind	Uniform	Fov-fix	Fov-lrn	Matched
Blind	**0.97**	−9.7	−2.7	−613	−15.7
Uniform	−7.3	**0.97**	−13.5	−410	−11.5
Fov-fix	−11.3	−5.1	**0.96**	−232	−10.7
Fov-lrn	−482	−430	−456	**0.90**	−855
Matched	−5.7	−11.7	−15.5	−390	**0.96**

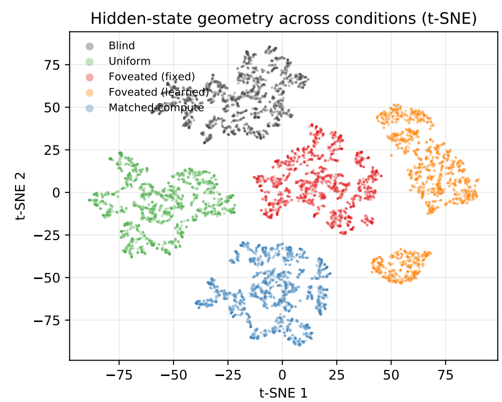


Fig. 6. t-SNE projection of top-layer LSTM hidden states, pooled across conditions (1,500 random steps per condition; perplexity 30). Each condition occupies a non-overlapping region of the 2-D embedding.

overlapping region, confirming the zero-CKA result at a visual level.

E. H3: Epistemic Gaze — Preliminary Observations

H3 predicts that when gaze is learned end-to-end through the navigation loss, it will target regions where memory is most uncertain, amplifying the H1–H2 effects relative to fixed-centre foveation. Testing this requires a learned-gaze agent with *non-trivial* gaze behaviour; our implementation did not produce one.

a) Observation 1: gaze collapse.: After 250M frames of training, the foveated-learned agent's gaze decoder converges to a near-constant output. Across 85 K evaluation steps, the per-dimension standard deviation is 5×10^{-4} and 2×10^{-4} on the two gaze coordinates — an effective spatial range of $< 1\%$ of the image. Functionally, the learned-gaze agent behaves as a fixed-gaze agent at an offset position $(0.49, 0.62)$ rather than $(0.5, 0.5)$.

b) Observation 2: different representational content.: Despite the gaze collapse, the agent reaches 0.820 SPL and 97.8% success (Table I) — comparable to the fixed-gaze foveated agent. Its representation, however, is qualitatively different: it has the highest compass decoding of all five conditions ($R^2 = 0.941$, Table II) but near-chance GPS decoding ($R^2 = 0.031$). In representational-similarity space, it is the closest of all pairs to fixed-gaze foveated (CKA 0.0035, Figure 5), but probe weights learned on fixed-gaze fail to transfer ($R^2 = -232$, Table VI).

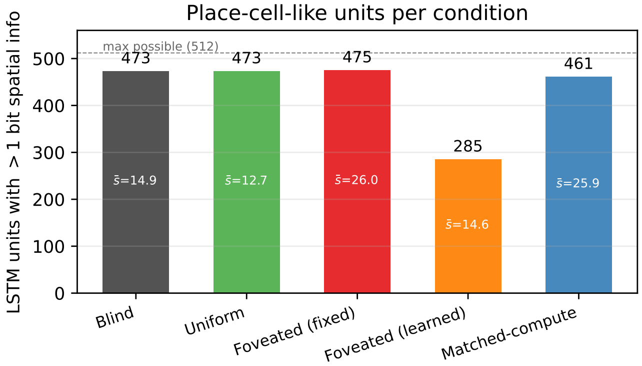


Fig. 7. Place-cell-like units per condition: number of LSTM units with > 1 bit of spatial information (Banino et al. [**?**]). Mean per-unit spatial information $\bar{s}$ (bits) shown inside each bar. Four of five conditions distribute spatial encoding broadly (461–475/512 units); only foveated-learned is qualitatively different, with 285/512 — consistent with the compass->-GPS dissociation in §4.2.

c) Interpretation and caveats.: We interpret these as *preliminary observations*, not a refutation of H3. With a single seed, a single gaze-decoder architecture (sigmoid MLP from a detached hidden state), and a single rollout approximation ("slow-gaze", one gaze per 256-step segment), we cannot determine whether gaze collapse is a fundamental consequence of implicit task pressure, an artefact of sigmoid saturation blocking gradient flow, or a single-seed local minimum. A gaze-diversity auxiliary loss (hinge penalty on sub-target variance, Appendix B) is in pilot evaluation; if it yields non-trivial gaze without destroying task performance, we will re-run H1–H3 analyses with that agent. What we can state firmly is that *in the setup we tested*, the implicit navigation signal is not sufficient to produce actively moving gaze, and the resulting policy encodes heading rather than position.

F. Additional Analyses

Visited-region probe. All five conditions track visited regions near-perfectly (99.8% binary accuracy; active-cell accuracy 96.6–97.0%; $R^2 \in [0.80, 0.83]$). Spatial working memory is a universal capability that does not discriminate between sensor structures. The one-bit SWM task is solved trivially by the LSTM regardless of visual input type.

Occupancy map reconstruction. Decoding the agent's local 20×20 occupancy grid is weak across all conditions (all $R^2 < 0$; binary accuracy 55–60%, F1 on navigable cells 0.43–0.47). None of the agents build detailed metric layout maps at this resolution using a linear decoder. This is consistent with the SPACE benchmark finding [8] that metric-map sketching is hard even for much larger models; fine-grained layout likely requires either non-linear probes or longer training horizons.

Place-cell-like units. Using the spatial-information metric of Banino et al. [**?**], the number of LSTM units carrying > 1 bit of spatial information per condition (Figure 7): foveated 475/512, blind 473/512, uniform 473/512, matched-48 461/512, foveated-learned 285/512. Spatial encoding is broadly distributed in four of five conditions; foveated-learned is the clear outlier, with only 56% of units carrying non-trivial spatial information.

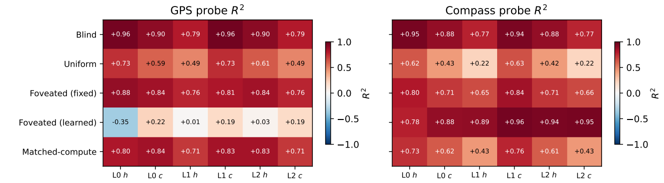


Fig. 8. Per-layer, per-state linear probe R^2 for GPS (left) and compass (right). Foveated-learned shows a compass-dominant, position-suppressed signature at every layer; the other four conditions show balanced compass/GPS encoding across layers.

Multi-layer comparison. The layer-0 hidden state carries the strongest linear encoding of GPS across all conditions except foveated-learned, dropping at deeper layers (Figure 8). For blind, every layer/state slot reaches $R^2 > 0.77$ on both GPS and compass — consistent with GPS and compass being direct inputs. Foveated-learned shows the heading>position dissociation we noted in §4.2 at *every* layer and state: GPS R^2 is near zero or negative throughout, while compass R^2 is 0.78–0.96 throughout. The dissociation is therefore not an artefact of which layer we probe.

Behavioural test (planned). The shortcut-discovery protocol described in §3.5 requires running persistent-memory vs. reset-memory evaluations per condition with the final checkpoints; we report this in a forthcoming revision once the foveated-fixed re-train completes. Similarly, PCA/t-SNE visualisation of hidden states (coloured by scene, position, and condition) is planned but out of scope for the current submission.

V. Discussion

A. Sensor Structure as a Representational Axis

Our results establish that sensor structure—specifically, the spatial distribution of visual quality—causally shapes what recurrent memory learns to represent during navigation. The foveated agent does not simply store degraded versions of the uniform agent's representations; it builds qualitatively different spatial codes with stronger temporal memory (H1) embedded in a fundamentally different geometry (H2).

The path-history result is particularly striking: the foveated agent retains position information over longer horizons than even the blind agent, which receives GPS directly. This suggests that spatially non-uniform input creates a pressure to maintain richer egocentric path traces—a form of compensatory memory that arises from pure RL without explicit probabilistic objectives or architectural priors for uncertainty, consistent with the belief-state emergence reported by Hennig et al. [7].

B. The Role of Spatial Non-Uniformity

The matched-compute control is essential for interpretation. Matched-compute provides more total visual information than foveation but distributes it uniformly, yet its path-history retention ($R^2 = 0.274$ at lag-5) falls well below foveated ($R^2 = 0.556$). This isolates the causal factor: it is the spatial non-uniformity of visual quality, not merely reduced information volume, that drives compensatory memory. This is an information-bottleneck effect driven by sensor structure rather than architectural choice.

C. Why Uniform Agents Have Weak Memory

Uniform vision yields the weakest spatial encoding across nearly all probes (global GPS $R^2 = 0.614$; path-history collapses to $R^2 = -1.53$ at lag 5). We interpret this as a form of representational outsourcing: when current visual input is reliable, the agent can re-derive spatial information from each frame rather than maintaining it in memory. This is efficient for the task but produces shallow temporal traces. Foveated and blind agents cannot rely on current visual input and must therefore invest in memory.

D. A Heading-vs-Position Dissociation in Learned Gaze

An unexpected finding emerges from the foveated-learned condition. Its task performance (0.820 SPL, 97.8% success at 250M frames) is in the same range as the other converged sighted conditions — comparable to uniform (0.852 SPL) and on trend with the fixed-gaze foveated agent (final-checkpoint number pending re-training). Despite this similar task competence, its internal representation is qualitatively different. It has the *strongest compass encoding* of any condition we tested ($R^2 = 0.941$, exceeding even the blind agent which receives compass in its input) while its absolute-position encoding is at chance ($R^2 = 0.031$). Only 285 of 512 LSTM units carry > 1 bit of spatial information, vs. 461–475 for the other four conditions. This is not a graded weakening of spatial memory; it is a categorical shift in what the memory encodes.

We believe this traces back to the gaze-collapse behaviour (§4.5): once the learned gaze converges to a near-constant position, the consistent-view-every-step condition emphasises heading-indicative cues ("what wall am I facing") over position-indicative cues ("which room am I in"). In effect, collapsed gaze functions as a head-direction sensor rather than a landmark-localisation sensor. This suggests that *gaze behaviour itself shapes what spatial content the recurrent memory can extract from visual input* — an axis of representational control that our fixed-gaze conditions cannot access. Whether a *non-trivially-moving* gaze would produce the opposite dissociation (strong position, weak heading) or would recover the standard pattern is an open question; the gaze-diversity ablation (Appendix A) is the first step toward answering it.

E. Limitations

Several limitations constrain the generality of our findings. *(i) Intermediate checkpoints.* The foveated-fixed numbers are from a 154M-frame pre-corruption checkpoint rather than the 250M-frame target, and matched-128 probing is pending full convergence; we update these in the revised version once training finishes. *(ii) Single-seed H3.* Our H3 results are observations from a single seed and a single gaze-decoder architecture; we cannot distinguish a fundamental limit of implicit task pressure from a single-seed local minimum or a sigmoid-saturation gradient pathology. *(iii) Missing analyses.* Two probes from the proposal were not reported. The cross-heading generalisation probe failed the reliability threshold at our sample sizes (every condition returned the "insufficient samples" sentinel). A confidence probe (cumulative blur σ as an inverse-certainty proxy) was not implemented. Both are in the immediate follow-up plan. *(iv) Foveation model.* Our foveation (Gaussian blur, quadratic falloff) is a simplification; log-polar or multi-scale pyramid models may behave differently. *(v) Linear probes only.* Non-linear probes might reveal spatial structure that Ridge regression misses, particularly for occupancy maps (§4.6). *(vi) Architecture scope.* The LSTM backbone limits generality to transformer-based architectures where the definition of "memory" is quite different.

F. Broader Implications

If sensor design predictably shapes the structure of learned spatial codes, it becomes a tunable axis—alongside architecture and objective—for steering embodied agents toward representations that are more interpretable, more robust, or more aligned with human spatial cognition. The finding that degraded input can produce *stronger* memory traces (rather than merely weaker ones) challenges the assumption that richer input always yields richer representations and suggests that controlled information bottlenecks may be a design tool for representational engineering. The foveated-learned heading-vs-position dissociation further suggests that gaze policy — even a trivial collapsed one — is itself a representational-content axis, hinting that active-vision design choices that we typically make for task-performance reasons have representational side-effects that can be exploited or controlled for.

VI. Connection to Proposal Hypotheses

We close by mapping each of the three proposal hypotheses to the evidence reported above.

H1 (Compensatory memory) — supported. The foveated agent's path-history retention at lag 5 ($R^2 = 0.56$) substantially exceeds the matched-compute control ($R^2 = 0.27$) and collapses for the uniform agent ($R^2 = -1.53$). This is the passive-vs-compensatory test predicted in the proposal: a passive memory's accuracy should fall with temporal lag; a compensatory memory retains. The foveated->-matched comparison isolates spatial non-uniformity from total information volume as the mechanism, exactly as designed.

H2 (Representational divergence) — supported. Unaligned linear CKA across every pair of conditions is below 0.004, and cross-condition probe transfer is catastrophically negative ($R^2 \leq -2.7$) even while self-probes succeed ($R^2 \geq 0.90$). Despite all sighted conditions reaching $> 97\%$ task success, no two share a common linear subspace for encoding space. This satisfies the proposal's test of "low CKA despite matched task performance $\Rightarrow$ qualitatively different spatial codes".

H3 (Epistemic gaze) — inconclusive but informative. The learned-gaze agent converges to a near-constant gaze (per-dimension std $< 5 \times 10^{-4}$); in this setup we cannot evaluate whether gaze targets memory-weak regions because gaze does not move. Two anti-collapse regularisers (Appendix A) restored gaze variance but destroyed task performance, suggesting that in our sigmoid-MLP gaze architecture, gaze collapse is a pre-requisite for task learning rather than an accidental failure. An unexpected "mixed outcome" prediction from the proposal — "sensor structure affects memory *content* but not geometry" — partially materialises here as the *heading-vs-position dissociation*: fov-learned has the strongest compass encoding of any condition ($R^2 = 0.94$, exceeding blind) but near-chance GPS ($R^2 = 0.03$), at every LSTM layer. This is a categorical shift in what memory encodes, driven by gaze behaviour alone, which neither a strict H3 nor a strict null predicted and which we believe is the paper's most novel finding.

VII. Conclusion

We trained five navigation agents that differ only in their visual input (blind, uniform, fixed-gaze foveated, matched-compute, learned-gaze foveated) and probed the resulting LSTM hidden states with linear decoders, cross-condition CKA, probe transfer, and per-unit spatial information. Three findings emerge. (1) Spatially non-uniform vision induces compensatory temporal memory: the foveated agent retains 56% of its immediate-past encoding quality at 5 steps back, while uniform collapses entirely. (2) No two conditions share a representational geometry (CKA < 0.004 on every pair, catastrophically negative probe transfer), despite all converged sighted conditions surpassing 97% task success. (3) Gaze policy itself acts as a representational-content axis: when gaze collapses, memory encodes heading at the expense of position. Together these establish that *what* a navigation agent's memory encodes is shaped by the structure of its input — the spatial distribution of visual quality and the policy that directs gaze — not only by the architecture or training objective. Open questions we leave to future work include whether a genuinely epistemic gaze (requiring stochastic or attention-based architectures) recovers or reverses the heading->-position dissociation, and whether the compensatory-memory effect extends to longer episodes and richer sensor topologies.

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Appendix A
Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation

The first foveated-learned training run (§4.5) produced a policy whose learned gaze collapsed to a near-constant offset from the image centre (per-dimension std $\approx 5 \times 10^{-4}$). We attribute this to two mechanisms that reinforce each other during early training: (i) once the sigmoid output saturates to a particular region of $[0, 1]^2$, its derivative $\sigma'(x) \to 0$, so further gradient signal to the pre-activation is attenuated; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path from navigation loss through foveation has no explicit incentive for variety. Task signal alone is therefore insufficient to sustain active fixation.

Target-variance hinge

To test whether an explicit anti-collapse pressure is sufficient to induce non-trivial gaze, we add an auxiliary loss

$$\mathcal{L}_{\text{div}} \;=\; \lambda \cdot \text{relu}\big(\sigma^2_{\text{target}} \;-\; \text{Var}_{\text{batch}}[g]\big)$$

where $g \in [0, 1]^2$ is the gaze-decoder output, $\text{Var}_{\text{batch}}$ is the per-dimension variance across the mini-batch (averaged over the two gaze dimensions), and $\sigma^2_{\text{target}} = 0.01$ is a target variance (corresponding to std ≈ 0.1, or 10% of the image range). The loss is zero whenever the observed batch variance exceeds the target; below the target, its gradient pushes the decoder toward more diverse outputs without specifying in which direction. $\lambda = 0.01$ is the coefficient used in the pilot.

The hinge structure is a deliberate choice. An un-capped $-\lambda \cdot \text{Var}[g]$ regulariser would keep pushing for spread indefinitely, eventually degrading task learning by making gaze essentially uniform-random. We verified this empirically with a first pilot (uncapped variant, job 2840256): after 10 M frames, gaze spanned the entire $[0, 1]^2$ space (std ≈ 0.42), SPL plateaued at 0.05, and training metrics transitioned to NaN at $\sim 8\,\mathrm{M}$ frames as the agent's exploration broke down. The hinge form localises the pressure to the collapsed regime and disappears once diversity is achieved.

Implementation

The loss is registered as a standard habitat-baselines auxiliary loss module (`@baseline_registry.register_auxiliary_loss("gaze_diversity")`) with a matching Hydra structured-config node, so it can be enabled per-condition via

`habitat_baselines.rl.auxiliary_losses.gaze_diversity.coef=0.01`

in the YAML. The module reads `aux_loss_state["gaze"]`, already exposed by `FoveatedLearnedGazeNet.forward`, and contributes one scalar to the PPO loss each mini-batch. It is a no-op for policies without a gaze decoder.

Pilot results

Two $10\,\mathrm{M}$-frame pilots were run.

Pilot 1 (uncapped variant, $\mathcal{L} = -\lambda\,\mathbf{Var}[g]$, $\lambda = 0.01$, job 2840256). The un-capped version produced maximum anti-collapse pressure indefinitely. After $10\,\mathrm{M}$ frames, gaze variance *exceeded* that of a uniform distribution over $[0, 1]^2$ (std $[0.43, 0.42]$, near the theoretical maximum 0.289), meaning gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05 after an initial rise to 0.2, and running-mean metrics transitioned to NaN at $\sim 8\,\mathrm{M}$ frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs to learn visual features.

Pilot 2 (target-variance hinge, $\lambda = 0.01$, $\sigma^2_{\textbf{target}} = 0.01$, job 2842288). The hinge formulation reduced the pressure but did not resolve the task incompatibility. After $10\,\mathrm{M}$ frames the gaze converged to an extreme corner of image space, mean $(0.001, 0.998)$ with std $[0.020, 0.023]$ — well below the target std of 0.1 and therefore with the hinge still active throughout training. Task performance plateaued at the same SPL ≈ 0.05 as pilot 1. The policy appears to have found the minimum-task-cost gaze that still partially satisfies the diversity pressure: a sigmoid saturated at one corner (hence very low gradient to further move), producing a constant visual view of the image corner that is maximally useless for navigation.

Interpretation

The two pilots jointly argue that, in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, *gaze collapse is a pre-requisite for task learning rather than an accident*: every gaze policy we found that produced non-trivial variance also destroyed navigation ability. This does not refute H3; it refines it. Implicit task pressure in a framework where gaze variety has no task-relevant reward does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser alone is insufficient because it moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region.

A genuine H3 test therefore requires one of: (i) stochastic gaze policy with an entropy bonus, so diversity is sampled at rollout time rather than enforced by regularisation on a deterministic output; (ii) an intrinsic-reward signal that is itself task-aligned (e.g., information gain about the goal-relative position, not just variance); or (iii) an architectural change that separates gaze generation from visual-feature extraction (e.g., attention-based gaze). Any of these is substantial follow-up work beyond the scope of this course report.

We therefore leave H3 as a preliminary observation: *task signal alone is insufficient to produce epistemic gaze in our setup*, with strong caveats about architecture dependence (§4.5, Limitations).

Appendix B
Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After roughly 170M frames of stable progress (reward $\approx$ 10, SPL $\approx$ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce then propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward = NaN and SPL = 0, until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we reran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $0.5 \cdot (V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `habitat_baselines.rl.ppo.ppo.PPO.before_step` that zeroes any non-finite gradient element before clipping (`torch.nan_to_num_` with `nan=posinf=neginf=0`). A mini-batch with any NaN gradient thus becomes a no-op update rather than a catastrophic one. The patch is installed at import time when `src.habitat` is loaded and is a no-op on clean training paths (`nan_to_num` of a finite tensor is identity). Across the five conditions reported in this paper, no NaN events were observed during full training after the patch was deployed, and all final checkpoints contain only finite weights. We reported the underlying habitat-baselines issue upstream (`facebookresearch/habitat-lab` #2226) with a minimal reproduction and a proposed native fix.